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[23CSE104]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2024

CYBERSECURITY LANDSCAPE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define CIA triad briefly.	L-1	CO1	**02**
	b.	Mention the disadvantage of fingerprint biometric.	L-2	CO2	**02**
	c.	Risk = ? **i.** Vulnerability + Threat **ii.** Virus + Malware **iii.** Vulnerability + Authentication **iv.** Phishing + Threat	L-2	CO3	**02**
	d.	**i.** Explain Dos attack?	L-2	CO4	**02**
	e.	Choose the correct order of RAR process. **i.** Identification → Analysis → Treatment → Evaluation **ii.** Identification → Evaluation → Analysis → Treatment **iii.** Identification → Analysis → Evaluation → Treatment **iv.** Analysis → Identification → Evaluation → Treatment	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Discuss Parkerian hexad.	L-2	CO1	**05**
		OR			
	b.	Tabulate the differences between vulnerability, threat, and risk.	L-2	CO1	**05**
3.	**a.**	Enlist and briefly explain the various attacks on authentication.	L-2	CO2	**05**
		OR			
	b.	What is transport layer security? Why is it important? Explain.	L-4	CO2	**05**
4.	**a.**	Discuss the key benefits of cyber threat management for an organization.	L-2	CO3	**05**
		OR			
	b.	Discuss about adversarial and non-adversarial threats.	L-2	CO3	**05**
5.	**a.**	How can an organization prevent denial of service attacks? Enlist and explain the important measures.	L-2	CO4	**05**
		OR			
	b.	Describe the lifecycle of a computer virus.	L-2	CO4	**05**
6.	**a.**	What is risk analysis? What are its major benefits? Discuss.	L-2	CO5	**05**
		OR			
	b.	Briefly explain the concept of cost/benefit analysis.	L-2	CO5	**05**

7.	a.	Create an exemplary dataset for spam email detection. What are the important factors/features responsible for an email to be considered as a spam? Justify your answer.	L-2	CO3	05
		OR			
	b.	Write an algorithm/program to implement two-factor authentication.	L-3	CO3	05
		PART-C	**06 X 10=60**		
8.	a.	What is cybersecurity? What are its various types and process of implementation?	L-3	CO1	10
		OR			
	b.	Suppose you are given the task to analyze the network of your organization and note down the vulnerabilities. Explain your action plan.	L-3	CO1	10
9.	a.	Discuss authentication and its types with the help of suitable examples.	L-2	CO2	10
		OR			
	b.	What is biometric authentication? Discuss along with its types, advantages, and disadvantages.	L-3	CO2	10
10.	a.	What do you mean by a threat in cybersecurity? What are the various types of threats popularly known? Discuss each with example.	L-2	CO3	10
		OR			
	b.	Discuss various roles of cybersecurity professionals.	L-2	CO3	10
11.	a.	Write a detailed note on denial of service attacks.	L-2	CO4	10
		OR			
	b.	Suppose you are a keynote speaker at a cybersecurity conference where you have to deliver a lecture on "How to protect yourself against cybercrime?" Discuss the important point you would cover in your lecture.	L-3	CO4	10
12.	a.	Explain risk analysis, its process, and major benefits in detail.	L-2	CO5	10
		OR			
	b.	Suppose you need to develop a risk management program in your organization? Elaborate the steps you would acquire.	L-3	CO5	10
13.	a.	How can the integration of continuous monitoring and adaptive risk assessment strategies enhance the resilience of an organization's cybersecurity posture in the face of rapidly evolving cyber threats?	L-3	CO3	10
		OR			
	b.	Perform the cost/benefit analysis for the installation of face biometric system.	L-3	CO5	10